**KIGALI INDEPENDENT UNIVERSITY ULK**

**School of Economics and Business Studies (EBS)**

**Department of Accounting & Finance**

**COST & MANAGEMENT ACCOUNTING /ASSIGNMENT 2/UNIT I**

**GROUP ASSIGNMENT**

**ACADEMIC YEAR 2025-2026 YEAR II /ACC & FIN/EVENING**

**SUBMISSION DATE: 24th/02/2026**

**Instructions:**

- ***Check well the list of group members and the accuracy of their student numbers before submitting your report, no complain will be received after submission.***
- ***Make sure the assignment has a clear HEADING [clear identification e.g. , department, module, group number, list of members etc]. The assignment without heading or clear identification will not be considered***
- ***Late submission will not be marked.***

# Part 1 : [From Chapter 1 to Chapter 3]

## Section 1.A: Introduction to Cost accounting

**N.B: attempt all questions in this section**

Q1.Cost accounting assists: (a) in controlling efficiency; (b) in pricing products; and (c) in providing a basis for operating policy. Amplify these points, giving reasons for your views.

Q2.What do you mean by elements of cost ? Discuss the various elements of cost

Q3.Define and explain the terms (a) cost centre (b) profit centre (c) cost object and (d) period cost

Q4.Financial accounting treats costs very broadly while the cost accounting does this in much greater detail" Explain this statement and state the limitations of financial accounting.

Q6. Explain each of the elements of the decision-making, planning and control process.

Q7. Explain how the business environment that businesses face has changed over the past decades and discuss how this has had an impact on cost and management accounting

Q8 Describe the different functions of management accounting.

## Section 1.B: Cost statement

**N.B: attempt 3 questions only**

### Question #1

### Q1.a :

i. Provide examples of each of the following: (a) direct labour, (b) indirect labour, (c) direct materials, (d) indirect materials, and (e) indirect expenses.

ii. Explain the meaning of the terms: (a) prime cost,(b) overheads, and (c) cost allocations.

iii. Distinguish between product costs and period costs.

### Q1.b :

For each of the following independent situations, calculate the missing values:

1. The Chico plant purchased $275,000 of direct materials during May. Beginning direct materials inventory was $16,000, and direct materials used in production were $200,000. What is ending direct materials inventory?
2. Landsman Company produced 10,000 units at an average cost of $6 each. The beginning inventory of finished goods was $3,510. (The average unit cost was $5.85.) Landsman sold 8,900 units. How many units remain in ending finished goods inventory?
3. Beginning WIP was $50,000, and ending WIP was $18,750. If total manufacturing costs were $93,000, what was the cost of goods manufactured?
4. If the conversion cost is $32 per unit, the prime cost is $19.50, and the manufacturing cost per unit is $39.50, what is the direct materials cost per unit?
5. Total manufacturing costs for April were $156,900. Prime cost was $90,000, and beginning WIP was $60,000. The cost of goods manufactured was $125,000. Calculate the cost of overhead for April and the cost of ending WIP.

### Question #2 :

BURUDANI Manufacturing Company is a producer of music compact discs (CDs) and . The following account balances are for the year ended December 31,2022.

| | |
|---|---|
| Administrative expenses | $ 60,000 |
| Depreciation expense – Manufacturing equipment | 50,000 |
| Direct labor | 468,000 |
| Manufacturing supplies expense | 40,000 |
| Indirect labor | 36,000 |
| Beginning inventories, January 1: | |
| Direct materials | 14,000 |
| Work in process | 20,000 |
| Finished goods | 128,000 |

| | |
|---|---|
| Ending inventories, December 31 | |
| Direct materials | 44,000 |
| Work in process | 56,000 |
| Finished goods | 92,000 |
| Direct materials purchases | 216,000 |
| Rent expense – Factory | 28,000 |
| Sales | 1,400,000 |
| Selling expense | 72,000 |
| Other manufacturing overhead | 126,000 |

Required :

**(i) Prepare a statement of cost of goods manufactured for Presley Manufacturing Company for the year.**

**(ii) Prepare an income statement for the year ended December 31**

**(iii) Prepare a Full cost statement**

**Question #3 :**

Mohamed LTD has supplied the following figures for the year ended 31 December 2016. Study the figures and select those required for the manufacturing account.

| | RWF |
|---|---|
| Factory machinery at cost | 80,000 |
| Maintenance of factory machinery | 18,000 |
| Factory buildings at cost | 230,000 |
| Purchases of raw materials | 220,000 |
| Opening stock of raw materials | 15,000 |
| Closing stock of raw materials | 12,500 |
| Returns of raw materials | 6,500 |
| Factory wages | 180,000 |
| Factory wages unpaid at 31 December | 5,600 |
| Factory rent and rates | 34,000 |
| Factory rent and rates due at 31 December | 2,570 |
| Office rent and rates | 12,500 |
| Office rent and rates due at 31 December | 1,040 |
| Factory light and heat | 2,600 |
| Office light and heat | 1,900 |
| Factory supervision | 25,000 |
| Office salaries | 53,400 |

Depreciation of factory machinery at 25% pa on the straight line method.

You are required to prepare a manufacturing account to show:

a) cost of raw materials consumed
b) direct wages
c) prime cost
d) total factory overheads
e) factory cost of production.

**Question #4**.

JC LTD produced 150,000 floor lamps during the past calendar year. JC Ltd had 2,500 floor lamps in finished goods inventory at the beginning of the year. At the end of the year, there were 11,500 floor lamps in finished goods inventory. The lamps sell for $50 each. JC's accounting records provide the following information for the past year:

| | |
|---|---|
| Purchases of direct materials | $1,675,000 |
| Direct materials inventory, January 1 | 380,000 |
| Direct materials inventory, December 31 | 327,000 |
| Direct labor | 2,000,000 |
| Indirect labor | 790,000 |
| Depreciation, factory building | 1,100,000 |
| Depreciation, factory equipment | 630,000 |
| Property taxes on the factory | 65,000 |
| Utilities, factory | 150,000 |
| Insurance on the factory | 200,000 |
| Research and development | 120,000 |
| Salary, sales supervisor | 85,000 |
| Commissions, salespersons | 370,000 |
| General administration | 390,000 |
| Work in process inventory, January 1 | 450,000 |
| Work in process inventory, December 31 | 750,000 |
| Finished goods inventory, January 1 | 107,500 |
| Finished goods inventory, December 31 | 489,000 |

***Required:***

1. **Prepare a cost of goods manufactured statement.**
2. **Prepare an income statement.**
3. **Compute the cost of producing one floor lamp last year.**

**Question #5.**

JOYCE Company makes products for sporting events. The following data are for the year ended December 31:

| | |
|---|---|
| Materials inventory, January 1 | $ 45,000 |
| Materials inventory, December 31 | 65,000 |
| Materials purchases | 175,000 |
| Direct labor | 225,000 |
| Work in process inventory, January 1 | 30,000 |
| Work in process inventory, December 31 | 40,000 |
| Manufacturing overhead | 130,000 |
| Finished goods inventory, January 1 | 80,000 |
| Finished goods inventory, December 31 | 140,000 |

**Required:**

**Prepare a Cost of Goods Manufactured Statement and the cost of goods sold**

**Question #6**.

KAPYEPYE Company's factory provided the following information for the last calendar year:
Beginning inventory:

| | |
|---|---|
| Direct materials | $49,300 |
| Work in process | 55,400 |
| Ending inventories: | |
| Direct materials | $20,000 |
| Work in process | 20,400 |

During the year, direct materials purchases amounted to $150,000, direct labor cost was $200,000, and overhead cost was $324,700. There were 100,000 units produced.

**Required:**

1. ***Calculate the total cost of direct materials used in production.***
2. ***Calculate the cost of goods manufactured. Calculate the unit manufacturing cost.***
3. ***Of the unit manufacturing cost calculated in Requirement 2, $1.70 is direct materials and $3.24 is overhead. What is the prime cost per unit? Conversion cost per unit?***

**Question #7**.

Ousmane MAHORO Company(OMC Ltd) produced 150,000 floor lamps during 2021. OMC Ltd had 2,500 floor lamps in finished goods inventory at the beginning of 2021. At the end of the year, there were 11,500 floor lamps in finished goods inventory. The lamps sell for $50 each. OMC Ltd 's accounting records provide the following information for the 2021:

| Items | RWF | Items | RWF |
|---|---|---|---|
| Direct materials inventory, January 1 | 380,000 | Insurance on the factory | 200,000 |
| Direct materials inventory, December 31 | 327,000 | Research and development | 120,000 |
| Purchases of direct materials | 1,675,000 | Salary, sales supervisor | 85,000 |
| Direct labor | 2,000,000 | Commissions, salespersons | 370,000 |
| Indirect labor | 790,000 | General administration | 390,000 |
| Depreciation, factory building | 1,100,000 | Work in process inventory, January 1 | 450,000 |
| Depreciation, factory equipment | 630,000 | Work in process inventory, December 31 | 750,000 |
| Property taxes on the factory | 65,000 | Finished goods inventory, January 1 | 107,500 |
| Utilities, factory | 150,000 | Finished goods inventory, December 31 | 489,000 |

**Required:**

i. ***Prepare a statement of cost of goods manufactured.***
ii. ***Prepare an income statement for OMC Ltd***

**Question #8**.

JULIUS SIMBA company is a metal and woodcutting manufacturer, selling product to the home construction market. Consider the following data for 2009.

| | |
|---|---|
| Sandpaper | 2,000 |
| materials- handling costs | 70,000 |
| lubricants and coolants | 5,000 |
| Miscellaneous indirect manufacturing labor | 40,000 |
| Direct manufacturing labor | 300,000 |
| Direct materials inventory Jan,1.2009 | 40,000 |
| Direct materials inventory Dec.31,2009 | 50,000 |
| Finished goods inventory Jan.1,2009 | 100,000 |
| Finished goods inventory Dec.31,2009 | 150,000 |
| Work-in-Process Inventory Jan.1,2009 | 10,000 |
| work-in-Process Inventory Dec.31,2009 | 14,000 |
| Plant leasing costs | 54,000 |
| Depreciation- Plant equipment | 36,000 |
| Property taxes on plant equipment | 4,000 |
| Fire insurance on plant equipment | 3,000 |
| Direct materials purchased | 460,000 |
| Revenues | 1,400,000 |
| Marketing promotion | 60,000 |
| Marketing salaries | 100,000 |
| Distribution costs | 70,000 |
| Customer- service costs | 100,000 |

**Required :**

***Prepare an income statement and a statement of cost of goods manufactured.***

**Question #9**.

Selected account balances for the year ended December 31 are provided below for MADARINA wa Kourtchev Company:

| | |
|---|---|
| Selling and administrative salaries.... | RWF150,000 |
| Insurance, factory..... | RWF7,500 |
| Utilities, factory..... | RWF30,000 |
| Purchases of raw materials..... | RWF165,000 |
| Indirect labor ..... | RWF45,000 |
| Direct labor..... | ? |
| Advertising expense ..... | RWF76,000 |
| Cleaning supplies, factory ..... | RWF4,900 |
| Sales commissions ..... | RWF54,000 |

| | |
|---|---|
| Rent, factory building..... | RWF110,000 |
| Maintenance, factory..... | RWF22,500 |

Inventory balances at the beginning and end of the year were as follows:

| | *Beginning of the Year* | *End of the Year* |
|---|---|---|
| Raw materials..... | RWF35,000 | RWF20,000 |
| Work in process..... | ? | RWF27,500 |
| Finished goods..... | RWF45,000 | ? |

The total manufacturing costs for the year were RWF479,000; the goods available for sale totaled RWF520,000; and the cost of goods sold totaled RWF500,000.

***Required:***

***Prepare a statement of cost of goods manufactured and the cost of goods sold section of the company's income statement for the year.***

## Section 2: Overhead allocation

***Attempt 3 questions in this section***

### Questions# 1.

a) In a multi-department production situation, explain the following:
   i) The role of service departments.
   ii) The reasons why it is important to distribute service department costs to production departments.

b) Select and explain two methods of distributing service department cost in the case where sercice departments also provide services to each other.

c) The following extract of information is available concerning the four cost centres of EG Limited.

| | *Production cost centres* | | | *Service cost centre* |
|---|---|---|---|---|
| | *Machinery* | *Finishing* | *Packing* | *Canteen* |
| Number of direct employees | 7 | 6 | 2 | – |
| Number of indirect employees | 3 | 2 | 1 | 4 |
| Overhead allocated and apportioned | $28,500 | $18,300 | $8,960 | $8,400 |

The overhead cost of the canteen is to be re-apportioned to the production cost centres on the basis of the number of employees in each production cost centre.

**Required:** After the re-apportionment, what is the total overhead cost of the packing department, to the nearest $?

**Questions #2.**

POLE POLE Company has two support departments (Administration and Janitorial) and three producing departments (Fabricating, Assembly, and Finishing). Costs and activities are as follows:

| | **Administration** | **Janitorial** | **Fabricating** | **Assembly** | **Finishing** |
|---|---|---|---|---|---|
| Direct costs..... | $50,000 | $30,000 | $40,000 | $50,000 | $25,000 |
| Number of employees..... | | 10 | 30 | 40 | 20 |
| Square feet..... | 2,000 | | 10,000 | 28,000 | 15,000 |
| Direct labor hours ..... | | | 5,000 | 6,000 | 2,000 |

Administrative services are allocated based on the number of employees; janitorial services are allocated based on square footage. Overhead rates for the three producing departments are based on direct labor hours.

***Required:***

***Determine the overhead for the producing departments using an allocation method at your choice***

**Questions #3.**

A factory consists of two production cost centres (P and Q) and two service cost centres (X and Y). The total allocated and apportioned overhead for each is as follows:

| *P* | *Q* | *X* | *Y* |
|---|---|---|---|
| $95,000 | $82,000 | $46,000 | $30,000 |

It has been estimated that each service cost centre does work for the other cost centres in the following proportions:

| | *P* | *Q* | *X* | *Y* |
|---|---|---|---|---|
| Percentage of service cost centre X to | 40 | 40 | – | 20 |
| Percentage of service cost centre Y to | 30 | 60 | 10 | – |

After the reapportionment of service cost centre costs has been carried out using a method that fully recognises the reciprocal service arrangements in the factory.

**Required : *what is the total overhead for production cost centre P?***

**Questions #4.**

A factory has two service departments P and Q and three production departments A, B, and C. You are supplied with the following information:

| Particulars | Total | Production Departments | | | Service departments | |
|---|---|---|---|---|---|---|
| | | A | B | C | P | Q |
| Rent | 12,000 | 2,400 | 4,800 | 2,000 | 2,000 | 800 |
| Electricity | 4,000 | 800 | 2,000 | 500 | 400 | 300 |
| Indirect labour | 6,000 | 1,200 | 2,000 | 1,000 | 800 | 1,000 |
| Depreciation of machinery | 5,000 | 2,500 | 1,600 | 200 | 500 | 200 |
| Other expenses | 4,500 | 910 | 2,143 | 847 | 300 | 300 |
| Estimated working hours | | 1,000 | 2,000 | 1,400 | 4,000 | 2600 |

Expenses of Service Departments P and Q are apportioned as under:

| | *A* | *B* | *C* | *P* | *Q* |
|---|---|---|---|---|---|
| P | 30% | 40% | 20% | — | 10% |
| Q | 10% | 20% | 50% | 20% | — |

**Required: Show the apportionment of overheads under different methods of apportioning inter- service departments overhead**.

**Questions #5.**

i. VANG- VANG factory has two service departments P and Q and three production departments A, B, and C. You are supplied with the following information:

| | | Production Departments | | | Services Departments | |
|---|---|---|---|---|---|---|
| **Particulars** | **Total** | **A** | **B** | **C** | **P** | **Q** |
| Rent | 12,000 | 2,400 | 4,800 | 2,000 | 2,000 | 800 |
| Electricity | 4,000 | 800 | 2,000 | 500 | 400 | 300 |
| Indirect labour | 6,000 | 1,200 | 2,000 | 1,000 | 800 | 1,000 |
| Depreciation of machinery | 5,000 | 2,500 | 1,600 | 200 | 500 | 200 |
| Other expenses | 4,500 | 910 | 2,143 | 847 | 300 | 300 |

Expenses of Service Departments P and Q are apportioned as under:

| | *A* | *B* | *C* | *P* | *Q* |
|---|---|---|---|---|---|
| P | 30% | 40% | 20% | — | 10% |
| Q | 10% | 20% | 50% | 20% | — |

**Required: Prepare an overhead analysis Sheet for VANG -VANG Factory**

# Part 2: [Chapter 4- Chapter 5]

## Section 1: Inventory valuation and control

**Attempt 3 questions in this section**

**Questions #1.**

The following information was extracted from the books of Dingida Holdings regarding its stocks:

| | | |
|---|---|---|
| i. | Reorder quantity | 1,800 |
| ii. | Reorder period | 4 weeks |
| iii. | Maximum consumption | 450 units/week |
| iv. | Normal consumption | 300 units/week |
| v. | Minimum consumption | 150 units/week |
| Vi | Maximum reorder period | 5 weeks |
| Vii | Minimum reorder period | 3 weeks |

**Required**

Determine the following stock levels for Diginda Holdings:

i. **Re-order level**

ii. **Maximum stock level**

iii. **Minimum stock level**

**Questions #2.**

Vinvira uses 15,000 kgs of plastic each year in its production of plastic cups. The cost of placing an order is $10. The cost of holding one kg of plastic for one year is $0.30. Vinvira uses an average of 60 kgs of plastic per day. It takes five days to place and receive an order.

***Required:***

1. **Calculate the EOQ.**

2. **Calculate the annual ordering and carrying costs for the EOQ.**
3. **What is the reorder point?**

**Questions #3.**

AMAHORO company manufacturers 5,000 units of a product per month. The cost of placing an order is Rwf 100. The purchase price of the raw material is Rwf 10 per kg. The re-order period is 4 to 8 weeks. The consumption of raw materials varies from 100 kg. to 450 kg. per week. The average weekly consumption being 275 kg. The carrying cost of inventory is 20% per annum. Assuming 52 weeks in a year,

**Required to calculate**

i. **Re-order quantity;**
ii. **Maximum level;**
iii. **Minimum level;**
iv. **Average level.**

**Questions #4.**

PAMOJA Ltd. uses about 75,000 valves per year and the usage is fairly constant at 6,250 valves per month. The valve costs Rwf 1.50 per unit when bought in large quantities; and the carrying cost is estimated to be 20% of average inventory investment on an annual basis. The cost to place an order and process the delivery is RWF. 18. It takes 45 days to receive delivery from the date of an order and a safety stock of 3,250 valves is desired.

**You are required to determine**

(i) **The most economical order quantity and frequency of orders;**
(ii) **( the re-order point; and**
(iii) **(the most economical order quantity if the valves cost RWF 4.50 each instead of RWF 1.50 each**

**Questions #5**

Assume the following purchases were made in KAZINIKAZI Ltd

| Date of purchase | Units purchased | Price/unit |
|---|---|---|
| 1st January | 500 | 100 |
| 2nd January | 600 | 200 |
| 3rd January | 800 | 400 |

Units used on 4th January are 900. Determine the value/cost of units used by using FIFO, LIFO and weighted average.

**Required:**

***Determine the cost of units used and the value of the closing stocks using FIFO, LIFO and Weighted Average.***

**Questions #6**

Good Luck Co Ltd makes the following purchases and sales.

| | | |
|---|---|---|
| 1 January | Purchases | 4,000 units for $10,000 |
| 31 January | Purchases | 1,000 units for $2,000 |
| 15 February | Sales | 3,000 units for $13,000 |
| 28 February | Purchases | 1,500 units for $3,750 |
| 14 March | Sales | 500 units for $1,200 |

**Required:**

- **At 31 March, what is the closing inventory valuations using FIFO?**
- **At 31 March, what is the closing inventory valuations using LIFO**?

## Section 2: Labor Costing

**Attempt 3 questions**

**Questions 1.**

(i) Provide 2 advantages and 2 disadvantages of piece rate method in labour cost computation;
(ii) KORA NEZA Limited manufactures tables for export trade. Jobs are allocated to two operators, GALIBWA and Yvette with bonus paid for hours saved.
In the month of February 2016 GALIBWA made 186 units and Yvette made 210 units for which the time allowed of 30 standard minutes and 25 standard minutes per unit respectively was credited.
The basic wage rate was RWF.18 per hour for both employees. For every hour saved, a bonus was paid at 20% of the basic wage rate. Hours worked in excess were paid at the basic wage rate plus two thirds. Galibwa completed his job in 44 hours and Yvette completed her job in 39 hours. A basic week has 40 hours.

**Required:** ***For each operator, compute :***

i) ***The amount of bonus payable;***
ii) ***The total gross wage payable;***
iii) ***The wages cost per unit***

**Questions 2**.

KAGABO produced 600 units in a week but 40 units were rejected. Time allowed per unit is 6 minutes. No payment is made beyond the time allowed but the bonus is paid on accepted units only. No penalty is imposed on the rejected units. KAGABO produced 600 units in 48 hours. His wage rate per unit is $ 15. Calculate his total wage on the basis of Halsey Bonus scheme

**Questions 3**.

Employee A is a carpenter and normally works 36 hours per week. The standard rate of pay is $3.60 per hour. A premium of 50% of the basic hourly rate is paid for all overtime hours worked. During the last week of October, Employee A worked for 42 hours. The overtime hours worked were for the following reasons:

| | |
|---|---|
| Machine breakdown: | 4 hours |
| To complete a special job at the request of a customer: | 2 hours |

**Required:**

***How much of Employee A's earnings for the last week of October would have been treated as direct wages?***

**Questions 4.**

KORA NEZA Limited manufactures tables for export trade. Jobs are allocated to two operators, Jean Marie and Christian with bonus paid for hours saved.

In the month of February 2016 Jean Marie made 186 units and Christian made 210 units for which the time allowed of 30 standard minutes and 25 standard minutes per unit respectively was credited.
The basic wage rate was RWF.18 per hour for both employees. For every hour saved, a bonus was paid at 20% of the basic wage rate. Hours worked in excess were paid at the basic wage rate plus two thirds. Jean Marie completed his job in 44 hours and Christian completed his job in 39 hours.

A basic week has 40 hours.

**Required:**
***For each operator:***

iv) ***The amount of bonus payable;***
v) ***The total gross wage payable;***
vi) ***The wages cost per unit***

**Questions 5.**

Total output of AMANI for one week was 540 units. He was allowed 8 minutes per unit. He completed these units in 52 hours. His wage rate per hour is $ 50. Calculate AMANI's total wage according to:

a) Halsey scheme

b) Halsey-Weir scheme

***c)*** Rowan scheme

**==*End* ==**